Section A – TCP/IP Architecture and Application Layer Protocols

Question 1. TCP/IP Protocol Architecture and Industrial Communication

An industrial facility uses PLCs, HMIs, engineering workstations, and production servers from different manufacturers. These devices communicate through an Ethernet network using the TCP/IP protocol suite. Explain how TCP/IP enables interoperability between different industrial devices and analyze the roles of the Application, Transport, Internet, and Network Access Layers during communication between a PLC and a monitoring server.

Good Answer:

1. TCP/IP is a vendor-independent protocol suite that establishes standardized communication rules, allowing industrial devices with different hardware and operating systems to exchange information.
2. The Application Layer provides services such as web communication, file transfer, name resolution, and network management.
3. The Transport Layer uses protocols such as TCP and UDP to deliver information between applications, using port numbers to identify the communicating services.
4. The Internet Layer uses IP addressing and routing to forward packets between local and remote networks.
5. The Network Access Layer handles local frame transmission using Ethernet or Wi-Fi, including MAC addressing and physical network communication.

Lecture reference: Section 5.1 – TCP/IP Overview and Layers, Slides 3–8.

Question 2. TCP/IP Encapsulation and End-to-End Packet Delivery

An engineer accesses a web-based monitoring dashboard hosted on an industrial computer located on a different IP subnet. The request must travel through several network devices before reaching the server. Explain how TCP/IP encapsulation occurs when the client sends the request and how the receiving server reconstructs the information. Distinguish the roles of IP addresses, MAC addresses, and transport-layer port numbers.

Good Answer:

1. The Application Layer generates the HTTP or HTTPS request containing the information required by the monitoring application.
2. For a TCP-based web connection, the Transport Layer adds a TCP header containing source and destination port numbers together with information supporting sequencing and acknowledgements.
3. The Internet Layer adds source and destination IP addresses to form an IP packet that can be routed between different networks.
4. The Network Access Layer adds the source and destination MAC addresses to construct a local frame, with the destination MAC identifying the next hop when the server is remote.
5. At the destination, de-encapsulation removes the corresponding protocol information in reverse order, and the original application data is delivered to the web server.

Lecture reference: Section 5.1 – How TCP/IP Layers Work Together, Slides 9–10.

Question 3. HTTP Communication and Web Server Troubleshooting

A manufacturing plant uses a web-based HMI to display production information. During commissioning, an engineer observes different HTTP responses, including 200 OK, 404 Not Found, and 500 Internal Server Error. Explain how HTTP request-response communication operates and analyze the significance of these status codes. Discuss how the engineer can use the responses to identify whether the problem is related to the requested resource or the server.

Good Answer:

1. HTTP defines communication between a web client and a web server using requests and responses to exchange web resources.
2. A request such as GET /index.html HTTP/1.1 asks the server to retrieve the specified resource using the HTTP GET method.
3. The response 200 OK indicates that the server successfully processed the request and returned the requested information.
4. The response 404 Not Found indicates that the requested resource was not found, while 500 Internal Server Error indicates that the server encountered an internal processing problem.
5. By interpreting the response codes, the engineer can distinguish successful communication from resource-location problems and server-side failures, reducing unnecessary troubleshooting of unrelated network components.

Lecture reference: Section 5.2 – HTTP: Protocol of the World Wide Web, Slide 13.

Question 4. Email Protocol Selection for Industrial Alarm Systems

A manufacturing company uses an automated system to send maintenance alarms to engineers by email. Engineers access their messages from computers and mobile devices, and the system must ensure that mailbox information remains synchronized. Compare SMTP, POP3, and IMAP and explain which protocols should be used for sending alarm notifications and retrieving or synchronizing email. Identify relevant communication ports and discuss the advantages of the selected approach.

Good Answer:

1. SMTP is responsible for sending outgoing email and transferring messages between mail servers, making it appropriate for transmitting automated maintenance alarms.
2. Common SMTP ports include TCP 25 for server-to-server relay, TCP 587 for client submission, and TCP 465 for implicit TLS submission.
3. POP3 retrieves or downloads messages from an email server, traditionally using TCP 110 or TCP 995 for secure connections.
4. IMAP provides server-based mailbox management and synchronization, including message status, folders, and deletions, commonly using TCP 143 or secure TCP 993.
5. SMTP should be used to send industrial alarms, while IMAP is preferable for engineers accessing the same mailbox from multiple devices because it maintains consistent mailbox information.

Lecture reference: Section 5.2 – Email Protocols: SMTP, POP3, and IMAP, Slide 14.

Question 5. FTP, TFTP, and SFTP for Industrial Device Configuration

An automation engineer must transfer configuration files and firmware images between an engineering workstation, industrial Ethernet switches, and PLC-related devices. The engineer is considering FTP, TFTP, and SFTP. Compare these three file-transfer protocols in terms of transport method, functionality, security, and reliability. Recommend an appropriate protocol for transferring sensitive configuration files and explain why TFTP may still be used with industrial equipment.

Good Answer:

1. FTP provides file-transfer and file-management functions over TCP, traditionally using TCP port 21 for control communication and a separate data connection.
2. Standard FTP does not encrypt usernames, passwords, or transferred data, making it inappropriate for sensitive transfers across untrusted networks.
3. TFTP is a lightweight protocol using UDP, with initial requests typically directed to UDP port 69, and provides basic file-reading and file-writing operations.
4. Although UDP does not guarantee delivery, TFTP implements simple reliability through data blocks, acknowledgements, timeouts, and retransmissions, making it useful for embedded-device configuration and firmware transfers on controlled networks.
5. SFTP is preferable when confidentiality and secure authentication are required because it operates through SSH, while TFTP may remain necessary for industrial devices that support only simple file-transfer mechanisms.

Lecture reference: Section 5.2 – FTP and TFTP Protocols, Slides 15–16; SSH Protocol, Slide 19.

Question 6. Secure Remote Access to Industrial Computers

A factory maintenance engineer must remotely configure a Linux-based industrial server and troubleshoot a Windows engineering workstation. The facility currently uses Telnet and direct Internet exposure of Remote Desktop Protocol (RDP). Compare Telnet, SSH, and RDP in terms of communication method, security, and intended use. Recommend appropriate remote-access approaches and explain how the facility can reduce cybersecurity risks.

Good Answer:

1. Telnet provides remote command-line access using TCP port 23, but it lacks modern encryption and may expose usernames, passwords, and session information.
2. SSH provides secure command-line communication, normally through TCP port 22, with encryption, integrity protection, and authentication.
3. RDP provides remote graphical access to Windows computers and commonly uses TCP port 3389, with UDP 3389 also supported in modern implementations.
4. SSH is appropriate for administering the Linux industrial server, while RDP is appropriate when graphical interaction with the Windows workstation is required.
5. The company should avoid exposing RDP directly to the Internet and use protective measures such as VPN access, multifactor authentication, firewall restrictions, strong account policies, and regular security updates.

Lecture reference: Section 5.2 – Remote Desktop, Telnet, and SSH Protocols, Slides 18 –19.

Question 7. SNMP Monitoring and Industrial Network Diagnostics

A manufacturing facility operates several managed industrial Ethernet switches, servers, wireless access points, and uninterruptible power supplies. The maintenance team wants a centralized monitoring system that can display device status and automatically report network faults. Explain how SNMP can support centralized monitoring, describe the responsibilities of the SNMP manager, agent, and Management Information Base (MIB), and analyze how GET, SET, TRAP, and INFORM messages are used.

Good Answer:

1. SNMP is an application-layer network management protocol used to monitor and manage devices such as industrial switches, routers, servers, and UPS systems.
2. An SNMP manager or Network Management System (NMS) requests monitoring information and presents it through dashboards, alarms, and management applications.
3. An SNMP agent runs on the managed device and accesses information defined by its Management Information Base, including interface status, traffic counters, errors, CPU usage, and operating conditions.
4. GET retrieves device information, SET can modify permitted parameters, TRAP sends unsolicited notifications, and INFORM supports acknowledged notifications.
5. SNMP typically uses UDP port 161 for management requests and UDP port 162 for notifications, while SNMPv3 supports stronger security through authentication, integrity protection, and encryption.

Lecture reference: Section 5.2 – SNMP Protocols, Slides 20–21.

Question 8. DHCP Address Allocation and Industrial Network Configuration

A manufacturing facility contains 80 engineering workstations, 20 PLCs, and several industrial Ethernet switches. The network administrator plans to implement DHCP for automatic IP address allocation while ensuring that critical control devices maintain predictable addresses. Explain the DHCP DORA process and describe how DHCP scopes, exclusions, reservations, and scope options can be used to support reliable industrial network operation.

Good Answer:

1. DHCP automatically provides network settings such as IP addresses, subnet masks, default gateways, DNS servers, and other configuration parameters.
2. The DORA process consists of Discover, Offer, Request, and Acknowledge, allowing a client to request and obtain an available IP address from the DHCP server.
3. A DHCP scope defines the range of addresses available for dynamic assignment together with information such as the subnet mask and lease duration.
4. Exclusions prevent particular addresses from being assigned dynamically, while reservations associate specific clients with consistent DHCP-managed addresses.
5. The administrator can use DHCP for engineering workstations while protecting critical PLC and switch addresses through static configuration, exclusions, or reservations, reducing address conflicts and unexpected connectivity changes.

Lecture reference: Section 5.2 – DHCP, Scope Options, Exclusions, and Reservations, Slides 22–24.

Question 9. DHCP Lease Renewal and APIPA Troubleshooting

An engineering laptop receives an IPv4 address from a DHCP server with a lease duration of eight hours. The DHCP server becomes unavailable after the address has been assigned. Later, the laptop experiences a communication failure and eventually receives an address from the 169.254.0.0/16 range. Calculate the normal T1 and T2 lease-renewal times, explain what happens when the DHCP server remains unavailable, and identify the engineering checks required to resolve the problem.

Good Answer:

1. DHCP normally begins its T1 renewal process at approximately 50% of the lease duration, so an eight-hour lease reaches T1 after four hours.
2. The T2 rebinding process normally begins at approximately 87.5% of the lease duration, corresponding to seven hours for this lease.
3. During T1 the client attempts renewal with the original DHCP server, while during T2 it attempts to obtain renewal from any available DHCP server.
4. If the lease expires without valid renewal, Windows may use an IPv4 link-local address from 169.254.0.0/16, which normally permits communication only on the local network link.
5. The engineer should investigate DHCP server availability, cabling, switch configuration, VLAN membership, DHCP relay operation, and firewall or network-path restrictions.

Lecture reference: Section 5.2 – DHCP Lease Process and Technical Details, Slides 25 –26.

Question 10. DNS Hierarchy and Name Resolution Across Networks

An industrial monitoring application connects to a production server using its domain name rather than a numerical IP address. When the operator enters the server name, the local DNS resolver does not have a cached result and must obtain the required address information. Explain how hierarchical DNS resolution works and analyze the responsibilities of the recursive resolver, root servers, top-level domain servers, and authoritative DNS servers. Discuss how caching improves communication performance.

Good Answer:

1. DNS translates human-readable domain names into information such as IP addresses, allowing applications to locate network services without requiring users to memorize numeric addresses.
2. The client's recursive resolver receives the query and attempts to obtain an answer using cached information or other DNS infrastructure.
3. If the information is unavailable locally, resolution can involve root DNS servers that direct the resolver toward the appropriate top-level domain servers.
4. The top-level domain infrastructure identifies the authoritative DNS servers responsible for the requested domain, and an authoritative server provides the relevant DNS record.
5. The resolver can temporarily cache the result according to its Time To Live (TTL), reducing repeated lookup delays and unnecessary requests to other DNS servers.

Lecture reference: Section 5.2 – DNS Protocol, Hierarchy, Servers, Root Hints, and Resolver Caching, Slides 27–30 and 35–36.

Section B – Transport Layer Protocols and Communication Reliability

Question 11. Transport-Layer Ports, Sockets, and Industrial Firewall Configuration

An industrial monitoring server provides web access to several engineering workstations. A client at 192.168.1.25 opens a connection from source port 53000 to a server at 192.168.1.10 listening on TCP port 80. Explain how ports and sockets identify communication endpoints, describe the TCP connection's four-tuple, and analyze why transport-layer port information is important for simultaneous client connections and firewall security.

Good Answer:

1. An IP address identifies a network interface, while a transport-layer port number identifies an application or service operating on that device.
2. A socket is a software communication endpoint identified by an IP address, port number, and transport protocol.
3. The TCP connection is identified by the four-tuple: source IP 192.168.1.25, source port 53000, destination IP 192.168.1.10, and destination port 80.
4. Multiple clients can connect to the same server port because their source IP addresses or source ports distinguish their individual connections.
5. Firewalls can control access using IP addresses, transport protocols, port numbers, and connection state, so unnecessary services and ports should be restricted to reduce the network's attack surface.

Lecture reference: Section 5.3 – Port Numbers, Port Security, Port Ranges, and Sockets, Slides 37–40.

Question 12. UDP Header Analysis and Data Length Calculation

An engineer captures a UDP datagram transmitted by an industrial device and records the following 8-byte UDP header in hexadecimal: CB84 000D 001C 0001. The UDP header contains four 16-bit fields: source port, destination port, total length, and checksum. Decode the header, determine the source port, destination port, UDP datagram length, and application-data length. Explain the role of the checksum and the limitations of UDP reliability.

Good Answer:

1. The source port field is CB84 in hexadecimal, corresponding to decimal port 52,100.
2. The destination port field is 000D, corresponding to decimal port 13.
3. The UDP length field is 001C, or 28 bytes, representing the combined UDP header and application payload.
4. Since the UDP header occupies eight bytes, the application-data length is \(28-8=20\) bytes, while the checksum field contains the value 0001.
5. The UDP checksum supports error detection but not error correction, and UDP does not provide built-in delivery acknowledgements, ordered delivery, or retransmission.

Lecture reference: Section 5.3 – UDP Protocol and UDP Header Structure, Slides 42–43.

Question 13. TCP vs. UDP for Industrial Data Transmission

A manufacturing plant uses one network application to transfer production reports to a central database and another application to send frequent, time-sensitive process measurements to a monitoring system. Compare TCP and UDP in terms of connection establishment, reliability, overhead, data ordering, and application suitability. Recommend an appropriate transport protocol for each application and justify your selections.

Good Answer:

1. TCP is connection-oriented and provides reliable, ordered byte-stream communication using acknowledgements, retransmissions, and sequence numbers.
2. UDP is connectionless and transmits individual datagrams without a connection-establishment handshake or built-in guarantees of delivery and ordering.
3. TCP includes mechanisms for flow and congestion control, while UDP has a smaller eight-byte header and lower protocol overhead.
4. TCP is suitable for transferring production reports because missing or incorrectly ordered information could compromise the completeness of stored records.
5. UDP may be appropriate for frequent process measurements where lower overhead and timely updates are priorities, provided the application can tolerate or manage dropped data.

Lecture reference: Section 5.3 – TCP and UDP Characteristics and Applications, Slides 41–43 and 47.

Question 14. TCP Sequence Numbers, Acknowledgements, and Loss Recovery

A production server sends 400 bytes of application data to an industrial workstation, beginning with TCP sequence number 1000. The workstation receives these bytes successfully, but a later segment is lost while additional data continues to arrive. Determine the expected acknowledgement number after the first segment, explain how cumulative acknowledgements identify missing data, and analyze how TCP retransmission, flow control, and congestion control support reliable communication.

Good Answer:

1. TCP assigns sequence numbers to transmitted bytes, so a 400-byte segment starting at sequence number 1000 contains bytes numbered 1000 through 1399.
2. After receiving the segment successfully, the workstation returns acknowledgement number 1400, indicating that byte 1400 is expected next.
3. If byte 1400 or a segment beginning there is missing, the receiver may continue sending cumulative acknowledgement number 1400 even if some later data arrives.
4. TCP can recover missing information through retransmission timeouts, duplicate acknowledgements, fast retransmission, and selective acknowledgements when supported.
5. Flow control prevents the sender from overwhelming the receiver's available buffer space, while congestion control adjusts transmission behavior to avoid placing excessive traffic on the network.

Lecture reference: Section 5.3 – TCP Characteristics, Sequence Numbers, Acknowledgements, and Loss Recovery, Slides 44 and 46.

Question 15. QUIC, TLS, and Modern Secure Web Communication

A manufacturing company is upgrading its browser-based production dashboard from HTTP/2 to HTTP/3. The engineering team wants to understand how the new communication architecture differs from traditional TCP-based web communication and how security is maintained. Compare traditional TCP-based HTTPS communication with HTTP/3 using QUIC. Explain the roles of UDP, QUIC, and TLS in providing reliable, secure, and efficient application communication.

Good Answer:

1. HTTP/1.1 and HTTP/2 commonly operate over TCP, which provides reliable, ordered communication between client and server applications.
2. HTTP/3 uses QUIC, a modern transport protocol that operates over UDP rather than directly over TCP.
3. QUIC adds reliability, retransmission, flow control, congestion control, and support for multiple independent communication streams.
4. TLS provides security functions including encryption, authentication, and data integrity, while QUIC integrates TLS 1.3 into its connection establishment.
5. HTTP/3 can improve web communication efficiency by combining secure connection establishment with multiplexed transport capabilities, although actual performance depends on network conditions and implementation.

Lecture reference: Section 5.3 – QUIC and TLS, Slide 48; TCP/IP Overview, Slides 6 and 10.

Question 16. SCTP Multi-Streaming and Multihoming for Reliable Communication

An industrial communication application exchanges equipment status, alarms, and control information between two geographically separated systems. The application must preserve individual message boundaries, support multiple independent streams, and maintain communication when one network path becomes unavailable. Evaluate the suitability of Stream Control Transmission Protocol (SCTP) for this application. Explain its message-oriented operation, multi-streaming, multihoming, association establishment, and reliability mechanisms.

Good Answer:

1. SCTP is a reliable, association-oriented transport protocol that preserves application-message boundaries rather than presenting information as a continuous byte stream.
2. SCTP multi-streaming allows several independent logical streams within one association, reducing the possibility that missing data in one stream blocks delivery in other streams.
3. SCTP multihoming allows an association to use multiple IP addresses and network paths, potentially maintaining communication through an alternative path if the primary path fails.
4. SCTP establishes an association through a four-way handshake consisting of INIT, INIT ACK, COOKIE ECHO, and COOKIE ACK, with a state cookie helping reduce resource-exhaustion attacks.
5. SCTP uses mechanisms such as selective acknowledgements, retransmission, flow control, and congestion control to provide reliable communication, making it suitable where message-oriented transport and network redundancy are important.

Lecture reference: Section 5.3 – SCTP Protocol, Multihoming, Handshake, and Applications, Slides 49–51.

Section C – Internetwork Layer Protocols and Routing

Question 17. ARP, IP Addressing, and Packet Forwarding Between Networks

An engineering workstation with IP address 192.168.1.10/24 must communicate with a production database server at 10.20.30.40. The workstation uses 192.168.1.1 as its default gateway. Explain how the workstation determines that the server is on a remote network, describe the role of ARP in preparing the first Ethernet frame, and analyze how IP and MAC addresses are used during routing.

Good Answer:

1. The workstation applies its subnet mask and determines that its local network is 192.168.1.0/24, while the destination server belongs to a different IP network.
2. Because the destination is remote, the workstation forwards the packet toward its configured default gateway at 192.168.1.1.
3. If the gateway's MAC address is not already cached, the workstation sends a local ARP broadcast request and receives the corresponding MAC address in an ARP reply.
4. The Ethernet frame uses the workstation's source MAC address and the gateway's destination MAC address, while the IP packet retains the database server's destination IP address 10.20.30.40.
5. Each router forwards the IP packet using its routing table and constructs a new local frame for the next link, while IPv6 networks use Neighbor Discovery with ICMPv6 rather than ARP.

Lecture reference: Section 5.4 – Routing, Address Resolution, ARP, and ARP Across Networks, Slides 52–57.

Question 18. ICMP, Ping, and Traceroute for Industrial Network Troubleshooting

A PLC can communicate with devices on its local network but intermittently fails to exchange data with a remote monitoring server. An engineer plans to use Ping and Traceroute/Tracert to investigate the communication path. Explain how ICMP supports network diagnostics and analyze how Ping and Traceroute can help identify communication problems. Discuss the significance of round-trip time, packet loss, TTL, and the limitations of ICMP testing.

Good Answer:

1. ICMP is an Internet Layer protocol used alongside IP for network diagnostics, error reporting, reachability testing, and path troubleshooting.
2. Ping sends ICMP Echo Request messages and examines Echo Replies to determine whether a device responds and to estimate round-trip time and packet loss.
3. Traceroute or Tracert uses mechanisms involving IPv4 Time To Live (TTL) or IPv6 Hop Limit values to investigate the routers along a communication path.
4. When a packet's TTL or Hop Limit reaches zero, a router may return an ICMP Time Exceeded message, helping identify the intermediate network hop.
5. A failed Ping does not necessarily mean that the destination is offline because ICMP traffic may be filtered by firewalls or affected by routing and congestion, and successful ICMP communication does not guarantee that an industrial application is functioning correctly.

Lecture reference: Section 5.4 – ICMP Protocol and Network Diagnostics, Slide 58.

Question 19. IPsec Security for Communication Between Industrial Facilities

A manufacturing company operates two production facilities that exchange production information over an external IP network. The engineering team needs to protect transmitted information against unauthorized interception and modification while allowing the internal networks to communicate. Evaluate how IPsec can provide secure communication between the facilities. Compare IPsec transport and tunnel modes, explain the functions of ESP and AH, and discuss the role of IKEv2 in establishing secure communication.

Good Answer:

1. IPsec provides security at the Internet Layer through mechanisms supporting confidentiality, data integrity, and authentication for IP communication.
2. IPsec transport mode protects the upper-layer payload while retaining the original IP header for routing, making it commonly associated with host-to-host communication.
3. IPsec tunnel mode protects the original IP packet and adds a new outer IP header, making it suitable for site-to-site VPN communication between industrial facilities.
4. Encapsulating Security Payload (ESP) can provide encryption, integrity, authentication, and anti-replay protection, while Authentication Header (AH) provides integrity and authentication without encryption.
5. IKEv2 negotiates security associations, cryptographic parameters, and keys between the communicating peers, supporting the establishment of protected IPsec communication.

Lecture reference: Section 5.4 – IPsec Protocol, Security Services, Modes, and IKEv2, Slides 59–60.

Question 20. Distance Vector Routing, RIP, and Routing-Loop Prevention

An industrial organization operates several connected IP networks using routers that exchange routing information. The engineering team is evaluating a distance-vector routing approach based on RIP. During testing, the failure of one network connection results in outdated route information and repeated packet forwarding between routers. Explain how distance-vector routing works, describe how RIP uses hop count to select routes, and analyze the count-to-infinity problem. Discuss how split horizon and poisoned reverse help reduce routing loops.

Good Answer:

1. Distance-vector routing allows routers to learn destination information from neighboring routers and calculate routes using the advertised distance together with the cost of reaching the neighbor.
2. RIP uses router hop count as its routing metric, normally preferring a destination route that requires fewer hops.
3. RIP considers routes with metrics from 1 to 15 reachable, while a metric of 16 represents an unreachable destination.
4. The count-to-infinity problem occurs when routers continue advertising outdated routes through one another after a failure, creating possible routing loops and increasing route metrics.
5. Split horizon prevents a router from advertising a route back toward the neighbor from which it learned the route, while poisoned reverse explicitly advertises that route as unreachable through the returning path.

Lecture reference: Section 5.4 – Distance Vector Routing, Count to Infinity, Split Horizon, Poisoned Reverse, and RIP, Slides 63–66 and 75–76.

r Routing and BGP, Slides 71–74 and 80–85.